INCH-POUND

MIL-C-46344A(AR)
AMENDMENT 3
28 August 1990
SUPERSEDING
AMENDMENT 2
5 AUGUST 1982

MILITARY SPECIFICATION

COMPUTERS, BALLISTICS: M13A2, M13A1DE1, AND M13A4

This amendment forms a part of MIL-C-46344A(MU), dated 14 August 1970, and is approved for use by the U.S. Army Armament Munitions, and Chemical Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* Delete specification title in its entirety and substitute: "COMPUTERS, BALLISTICS: M13A2, M13A1DE1, and M13A4"

* Delete 1.1 in its entirety and substitute:

"1.1 Scope. This specification covers three types of instruments known as Ballistics Computers M13A2, M13A1DE1, and M13A4, which are used to receive the log of range data from a range finder calibrated for range measurement in meters and compute superelevation for various ammunitions and transmit same to a gun sighting system."

* Delete paragraph 2.1 and substitute:

"2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2)."

* Below MIL-STD-109, add:

"MIL-STD-45662 Calibration Systems Requirements

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094)

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation."

* Delete "U.S. Army, Frankford Arsenal" and substitute "ARDEC"

* Delete list of drawings and substitute:

"7660281	Operating Instructions	
8258567	Operating Instructions	
8565930	Test Fixture, Ballistic Computer, M13A2	
8585200	Fixture, Oscillating Test	
8621000	Computer, Ballistics: M13A2	
10555194	Computer, Ballistics: M13A1DE1	
P8621000	Packaging of Computer, Ballistics:	M13A2
P10527413	Packaging of Computer, Ballistics:	M13A4
P10555194	Packaging of Computer, Ballistics:	M13A1DE111

* Add Beneficial Comments block:

"

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, NJ 07806-5000, by using a self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

"

PAGE 2

* Delete "PACKAGING DATA SHEETS" and list of Packaging Data Sheets.

* Delete: "(Copies of specifications, standards, drawings and packaging. . . .)"

* Substitute: "(Copies of other Government documents, drawings and publications required by contractors, in connection with specific acquisition functions, should be obtained from the contracting activity or as directed by the contracting officer.)"

* Add:

"2.2 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained."

* 3.2, delete and substitute:

"3.2 Fabrication. The computers shall be manufactured in accordance with Drawing 8621000 (Ballistics Computer, M13A2, Drawing 10527413 (Ballistics Computer, M13A4), Drawing 10555194 (Ballistics Computer, M13A1DE1), and drawings pertaining thereto. "

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3.1.2, delete item a and reletter items b through e as a through d.

PAGE 3

3.4.2.1, add the following to Ammunition/Ranges list:

Ammunition	Ranges
"APFSDS-T, XM735	500, 1,000, 2,000, and 3,600 meters"

PAGE 4

3.5.1.2, delete TABLE I and substitute:

"TABLE I. Position of superelevation counter.

Output Shaft Position	120.96° ccw	1200 meter checkpoint 0°	152.64° cw	1R8.64° cw	1R296.64° cw
Tolerance	± 8°	± 3°	± 8°	± 8°	± 8°
Superelevation Counter (mil)	1.20	2.88	5.00	8.00	12.00

PAGE 5

Add to TABLE II below "APERS-T" entry:

"

APFSDS-T XM735	0R 126.72° CCW *1.12	0R 43.20° CCW *2.28	0R 8.64° CCW *2.76	0R 36.72° CW *3.39	0R 118.80° CW *4.53	0R 298.80° CW *7.03	1R 54° CW *8.63	1R 216.72° CW *10.89

"

PAGE 6

3.5.2, add the following to Ammunition\Ranges list:

Ammunition	Ranges
"APERS-T, XM494E3	500, 1,000, 2,000 and 3,600 meters
APFSDS-T, XM735	500, 1,000, 2,000, and 3,600 meters"

PAGE 8

Table V, delete Voltage column in its entirety and substitute:

"Voltage (± 0.5 V)
28 V
28 V
18 V
18 V"

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3.5.6, delete “M393 and ADERS-T followed by two spacers” and substitute "M393, APERS-T, XN494E3, and APFSDS-T, XM735, followed by one spacer”

PAGE 9

* Delete paragraph 4.1 in its entirety and substitute:

“4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein; unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification, where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.”

* Below paragraph 4.1, add:

“4.1.1 Responsiblity for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specifications shall become a part of the contractor’s overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.“

PAGE 12

Delete 4.5, 4.5.1, 4.5.1, 4.5.2, 4.5.2.1, and 4.5.2.2 in their entirety and substitute:

“4.5 Inspection equipment. Unless otherwise specified in the contract or purchase order, the contractor shall supply, maintain, and calibrate inspection equipment in accordance with the provisions of MIL-I-45607 and MIL-STD-45662. The contractor shall submit inspection equipment drawings for approval in accordance with the terms of the contract (see 6.3).”

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4.6.3.4.2, line 5, delete “1500” and substitute “1200”

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* Add parenthetical comment below "6. NOTES”:

“(This section contains information of a generic or explanatory nature that may be helpful, but is not mandatory)”

* 6.1, line 2, delete “105 MM Gun Tank: M60A1” and substitute “105 Gun Tank, M60A1 and M48A5"

* Below paragraph 6:2, add:

“6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjuction with the conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Ref. Para.	DID Number	DID Title	Suggested Tailoring
4.5	DI-R-1714	Quality Engineering Acceptance inspection Equipment Descriptive Documentation	See 6.3.1

The above DID was that cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.3.1 Contractor designs. Contractor designs are required for all inspection equipment for which Government designs are not specified and include commercial equipment which the Contractor purposes to use. (Commercial equipment is defined as unmodified equipment which is catalogued and available for purchase by the general public.) Contractor designs shall include appropriate operating instructions, calibration procedures, and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment or with Government designs, if not otherwise covered thereby. Designs shall be of the category and form (per DOD-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The item detail specification number and paragraph number from Section 4 shall be referenced on each contractor design together with the component or assembly drawing number, revision letter, and date to which the specific design applies.

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6.3.2 Submission of designs for approval. Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for aproval in accordance with the stipulations, time frame, and distribution specified in the Contract Data Requirements Lists (DD Form 1423). Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule, as stipulated in the contract. Submit designs as required to: Commander, US Army Armament Research, Development, and Engineering Center, ATTN: AMSMC-QAF-I (D), Picatinny Arsenal, NJ 07806-5000.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament, Development, and Engineering Center (ARDEC) may also include drawings prepared by and identified as US Army Armament Research and Development Command (ARRADCOM), Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.5 Subject term (key words) listing.

Fire Control"

NOTE: The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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Preparing Activity:
Army - AR

(Project 1220-A397)

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